SET A

MakeUp Quiz

Name

ID

Section

A bank wants to decide whether to approve a loan (Yes/No) for customers based on the following categorical features:

- Age_Group: {Youth, Middle-aged, Senior}
- Income_Level: {High, Medium, Low}
- Student: {Yes, No}
- Credit_History: {Good, Bad}

Age_Group	Income_Level	Student	Credit_History	Approve_Loan
Youth	High	No	Good	No
Youth	High	Yes	Bad	No
Youth	Medium	Yes	Good	Yes
Youth	Medium	No	Bad	No
Senior	Low	Yes	Good	Yes
Senior	Low	No	Good	No
Senior	Medium	Yes	Bad	No
Senior	Medium	No	Bad	No
Middle-aged	High	Yes	Good	Yes
Middle-aged	High	No	Good	No
Middle-aged	Low	Yes	Bad	No
Middle-aged	Low	No	Good	Yes

1. Compute the entropy of the dataset with respect to the target Approve_Loan.
2. Calculate the **information gain** for each attribute and select the root node.
3. Construct a **decision tree** (ID3 algorithm) until all leaves are pure or no attributes remain.

4. **Using your tree, predict the loan decision for a new customer:**

    - Age_Group = Senior, Income_Level = Low, Student = Yes, Credit_History = Bad.

$Entropy(Approved\ loan) = -P_{Yes}\log_2 P_{Yes} - P_{No}\log_2 P_{No}$

$$= -\frac{4}{12}\log_2\frac{4}{12} - \frac{8}{12}\log_2\frac{8}{12}$$

$$= 0.918$$

∴ Age Group :→

$$Entropy(Youth) = -P(Yes|Youth)\log_2 P(Yes|Youth) - P(No|Youth)\log_2 P(No|Youth)$$

$$= -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4} = 0.81128$$

$$Entropy(Senior) = 0.81128$$

$$Entropy(Middle\ Aged) = 1$$

$$\therefore IG(Age) = 0.918 - \left(0.81128\times\frac{4}{12} + 0.81128\times\frac{4}{12} + 1\times\frac{4}{12}\right)$$

$$= 0.0438$$ ✓

∴ Income level :→

$$Entropy(High) = -P(Yes|High)\log_2 P(Yes|High) - P(No|High)\log_2 P(No|High)$$

$$= 0.81128$$

$$Entropy(Medium) = 0.81128$$

$$Entropy(Low) = 1$$

$$\therefore IG(Income\ level) = 0.0438$$ ✓

Student:

$Entropy(Yes) = 1$ $\quad$ $Entropy(No) = -\frac{1}{6}\log_2\frac{1}{6} - \frac{5}{6}\log_2\frac{5}{6}$

$$= 0.6500$$

$$\therefore IG = 0.093$$ ✓

Credit

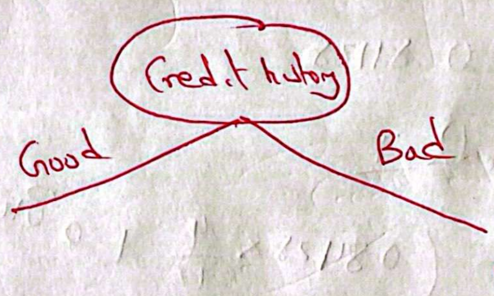
$\text{Entropy(Good)} = -\frac{4}{7}\log_2\frac{4}{7} - \frac{3}{7}\log_2\frac{3}{7} = 0.9852$

$\therefore \text{Entropy(Bad)} = 0$

$\therefore IG = 0.918 - \left(0.9852 \times \frac{7}{12}\right)$

$= \underline{0.343}$

$\therefore$ $\boxed{\text{IG Highest (Credit)}}$



* Now ~~considering credit~~ consider table with credit history Good.

$\therefore \text{Entropy(Target)} = -\frac{4}{7}\log_2\frac{4}{7} - \frac{3}{7}\log_2\frac{3}{7}$

$\underline{0.98}$

Age Group:

$\text{Entropy(Youth)} = 1$

$\text{Entropy(Middle Aged)} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$

$= 0.918$

$\text{Entropy(Senior)} = 1$

$IG(\text{Age Group}) = \underline{0.01514}$

Income level :-

$$Entropy(High) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$= 0.918$

$Entropy(M) = 0$

$Entropy(Low) = 0.918$

$$\therefore IG(Income\ level) = 0.98 - \left(0.918 \times \frac{3}{2} + 0 + 0.918 \times \frac{3}{2}\right)$$

$= 0.193$

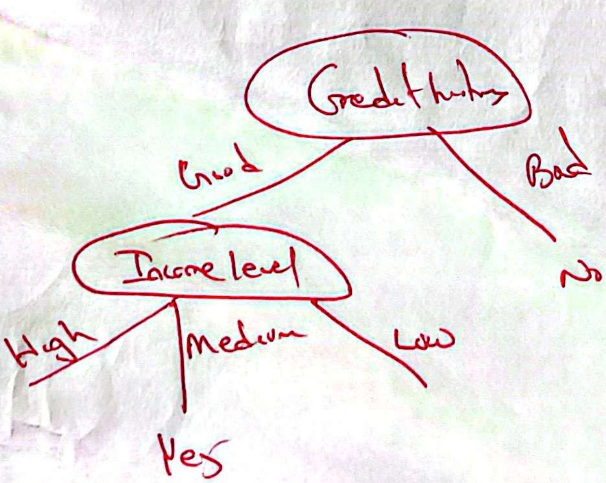
Student :-

$$Entropy(yes) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.918$$

$$Entropy(No) = -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4} = 0.8112$$

$$\therefore IG(student) = 0.98 - \left(0.918 \times \frac{3}{2} + 0.811 \times \frac{4}{7}\right)$$

0.123



∴ Now considering Income level high with Credit history Good

$$Entropy(Target) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$$

0.918

∴ Age Group :→

Entropy (youth) = 0

Entropy (middle Aged) = 1

∴ $IG = 0.918 - \left(1 \times \frac{2}{3}\right)$

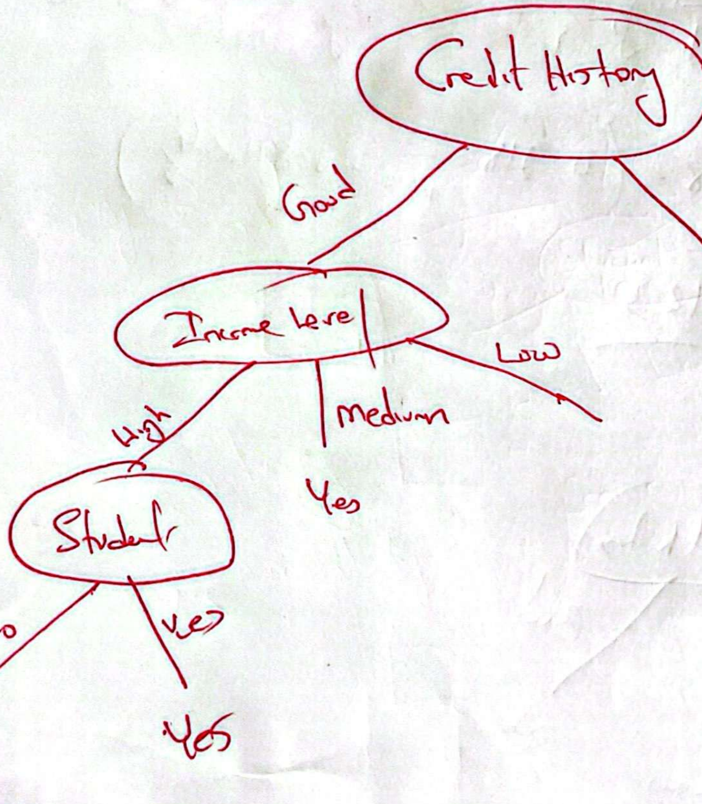
0.251

Student

Entropy (No) = 0

Entropy (Yes) = 0

∴ IG (Student) = 0.918



o Age_Group = Senior, Income_Level = Low, Student = Yes. Credit_History = Bad.

Now considering Income Level Low with credit history Good

Entropy (Target) = 0.918

Age Group:

Senior = 1.00

middle Aged = 0.

IG (Age Group) = 0.918 − (0.5H)

0.418

Student:

Yes = 0

No = 1.00

IG = 0.418

∴ So take anyone since equal IG